AI-Powered Recycling Bin: Waste Sorting with MobileNetV2

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***Abstract*—In this research, recyclable materials—plastic, paper, glass, metal, and general waste-are automatically categorized using an AI-powered recycling bin sorting system. The objective is to increase the effectiveness of trash management, support sustainable waste management practices, reduce human error, and accelerate the classification process. Using computer vision and machine learning techniques, the system identifies the type of waste and assigns it to the correct category in real-time. The prototype is evaluated based on accuracy and processing speed, demonstrating the potential of the system to enhance recycling efficiency and promote environmental sustainability in public and private environments.**

I. Introduction

A major obstacle to effective waste management is the growing amount of waste produced globally. Traditional manual sorting techniques are expensive, time-consuming, and error-prone, which may result in ineffective recycling procedures and contamination of recyclable materials. Advances in artificial intelligence (AI) and computer vision offer promising solutions that enable automated waste classification systems that can operate quickly and accurately.

This research proposes an AI-powered recycling bin sorting system capable of detecting and classifying common recyclable materials—plastic, paper, glass, metal, and general waste. By automating the sorting process, the system aims to reduce human error, increase efficiency, and improve sustainable waste management practices, thus contributing to environmental protection. The model is trained using the TrashNet dataset [1], which contains labeled images of common waste items. To ensure practical applicability, the performance of the system is further evaluated under real-world conditions, including variations in lighting, orientation, and partial occlusion [12].

Currently, smart recycling bins and AI-based waste management systems have made significant progress. Systems such as WasteNet [1] and DEEPBIN [6] can classify common waste items such as plastics, metals, and paper with high precision in controlled environments. Some smart bins now operate directly on edge devices like the Jetson Nano or low-power chips [2], allowing real-time classification without relying on cloud processing. Many prototypes also incorporate sensors, IoT connectivity, and reward mechanisms to encourage recycling [7,9]. However, most of these systems are still limited in scope—they are usually designed for specific types of waste, tested under ideal conditions, or deployed at a small scale. Large-scale integration with municipal waste management or handling of mixed and unpredictable waste streams remains a challenge. Despite these advances, there is still a need for AI systems that are flexible, cost-effective, and capable of performing reliably in messy real-world environments.



Fig. 1: The five waste categories

The ultimate goal of this research is to develop a system that not only performs well in controlled environments but also generalizes effectively to real-world scenarios, providing a scalable and reliable tool for smart waste management in both public and private settings.

II. Literature Review

Researchers are paying a lot of attention to how artificial intelligence can make waste management more efficient. Sorting trash by hand is slow, tiring, and often unsafe. The big hope is that AI can make the process faster and more reliable, while also cutting down on costs and mistakes. But how far have we

actually come, and what are the limits? Different studies give us some ideas.

White et al. [1] built a system called WasteNet, a CNN model that runs on small devices like the Jetson Nano. It can sort items such as paper, plastic, metal, glass, and cardboard right at the disposal point without needing cloud servers. That makes it much faster and more practical for "smart bin" designs. The problem is that it was tested on the TrashNet dataset, which is small and very controlled. Real waste is messier—dirty, folded, or under poor lighting. So, while WasteNet shows what's possible, it also reminds us that lab success doesn't always carry over to the real world.

Li and Grammenos [2] took things a step further by designing a smart bin that runs directly on edge devices. To make it safer, they added extra categories like "hand" and "empty," which help avoid accidents when people are near the bin. Their system worked on both a Jetson Nano and a much lower-power K210 chip, reaching about 95–96% accuracy. That's impressive considering the energy savings. Still, the study was mostly tested in tidy conditions, not in crowded or unpredictable public spaces.

Other researchers explored what happens when people and machines work together. Ramadurai and Jeong [3] studied how human involvement can improve a robot's performance. For example, if a person helps by moving an item or adjusting its position, the robot's accuracy jumps from around 33% to nearly 100%. That's a huge difference. It suggests that, for now, hybrid systems may be more realistic than fully independent ones. Cheng et al. [4] worked on a different angle: they built a robot in Japan that sorts beverage containers using suction-based grippers. It's fast and reduces the need for human labor, but it only works for PET and glass bottles. Expanding it to other types of waste will take more work.

Arthur et al. [5] looked at the bigger picture by reviewing smart bin prototypes that combine IoT, machine learning, and deep learning. Their review shows that while many ideas look good in the lab, very few have been deployed in real-world city systems. The main gaps are cost, scalability, and integration with municipal services. Song et al. [6] developed DEEPBIN, which connects AI sorting directly with IoT platforms for monitoring. This way, administrators can track performance in real time. The hardware is expensive, and rolling it out citywide is still a challenge.

There are also region-specific innovations. Atwah et al. [7] tested a smart bin in Saudi Arabia that uses AI sorting combined with a reward system to encourage people to recycle. The bin also monitors its fill level and condition in real time. While promising, it's still at the prototype stage, so we don't know how it would perform on a large scale. Moktar et al. [8] focused on medical waste and built a system using YOLO detection and IoT tracking. It can quickly identify gloves, masks, syringes, and other hazardous items. The main drawback is that it's limited to medical use and would need adjustments to handle normal municipal waste.

Alourani et al. [9] worked on a bin that combines cameras, sensors, and deep learning to improve real-time classification and collection scheduling. This fusion of sensor data makes the system more accurate, though it can also drive up costs and may struggle in cluttered environments. Fang et al. [10] shifted focus to plastic recycling, reviewing how AI can better separate different plastic types. This is important because plastic is one of the hardest materials to properly recycle. Their review points out that AI can boost accuracy and reduce contamination, but it mostly applies to plastics, not to mixed waste streams.

Some researchers have tried combining AI with other technologies. Alabdali et al. [11] introduced a system that uses AI for sorting while also storing data securely on blockchain. This makes the process transparent and tamper-proof, which can build trust in waste tracking. The downside is that the system needs specialized equipment and expertise, making it harder to scale in large cities. Ahmed et al. [12] trained a ResNet model on more than 15,000 images, classifying waste into 12 categories. It worked well even under changing lighting conditions, but it required heavy computing power, which isn't practical for cheap or lightweight bins.

Flexibility is another area of research. Vukicevic et al. [13] designed a vision system that can adapt to new waste items without starting training from scratch. That saves time in settings where waste streams constantly change. The tradeoff is that adding new categories still requires annotation work, and extreme variations can reduce accuracy. Sayem et al. [14] trained a model on 10,000 images to recognize 25 types of recyclable waste. The approach is strong, but the dataset is still small compared to the messy variety of real-world trash.

Finally, Babé et al. [15] explored the use of large vision models, often called foundation models, for waste classification. These models are trained on huge datasets and can adapt better to new conditions. The study showed that they outperform smaller models when used in new locations or times, making them more robust for real-world use. However, they require a lot of computing power and can be costly.

Despite systems such as WasteNet [1], DEEPBIN [6] and other smart bins displaying outstanding performance in controlled environments, artificial intelligence (AI) has demonstrated a significant guaranty to address these issues by making waste sorting faster, safer and more accurate [2,5]. However, the majority models are less effective in messy real-world scenarios, as they are trained on tiny or clean datasets and focus on particular types of waste, such as plastics [10] or medical products [8]. Few systems have been scaled or integrated with local trash management [5,7,9], and many rely on costly or powerful tools [12]. Another difficulty is flexibility, since the addition of new categories of waste can often require substantial human classification or retraining [13,14]. Accuracy can be increased through human-machine cooperation [3], but fully automated systems still have difficulty in complicated settings. Our approach seeks to fill these gaps by providing AI recycling solutions that are adaptable, affordable, able to handle mixed and unpredictable waste, and feasible for real-world implementation.

III. Proposed System Model

The proposed system integrates transfer learning, computer vision techniques, and deep-learning classification within a unified pipeline designed for accurate and real-time waste sorting. This section provides an overview of the algorithms used, the overall system workflow, and the evaluation measures applied to validate model performance.

A. Algorithms and Techniques Used

- **MobileNetV2 Transfer Learning**: Used as the base model for feature extraction.
- **Data Augmentation**: Enhances dataset diversity and improves robustness.
- **Softmax Classification Layer**: Produces class probabilities for the five waste categories.
- **Dropout Regularization**: Reduces overfitting and improves generalization.

B. System Pipeline

The complete workflow of the AI recycling bin sorting system consists of the following stages:

1) Collect images of waste items from a camera or an existing dataset.
2) Preprocess images: resizing, normalization, and augmentation.
3) Extract features using the pretrained MobileNetV2 model.
4) Classify items using custom dense layers.
5) Generate output: predicted waste category for sorting.

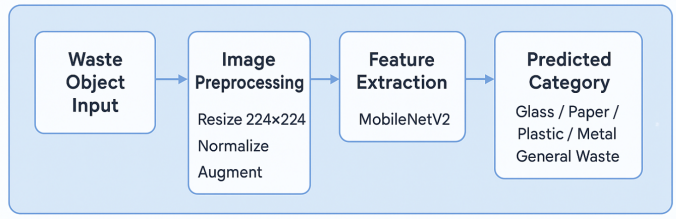


Fig. 2: System Pipeline of the AI Recycling Bin Sorting System

C. Evaluation Measures

To assess the reliability and real-world applicability of the system, multiple metrics and evaluation tests are applied.

1) Dataset-Based Accuracy: classification accuracy is first measured using a held-out validation set from TrashNet. This metric represents the proportion of correctly classified items across all five categories.
2) Inference Time: evaluates how quickly the system can classify each waste item. Low latency is essential for real-time operation in smart recycling bins deployed in high-traffic areas.
3) Robustness to Real-World Noise: additional testing is conducted using images outside the standard dataset. Tests include:
   - Rotated or partially occluded items
   - Low-resolution or blurry images
   - Poor lighting, shadows, or uneven illumination

   These tests ensure that the system remains reliable under practical, non-ideal conditions.
4) Confusion Matrix: generated to visualize where the model makes mistakes, such as confusing plastic with glass, helping to identify areas for improvement.



Fig. 3: Concept design of the Future Smart Bin integrating AI-based waste sorting.

IV. Methodology

The AI recycling bin sorting system is designed to automatically classify five types of waste: glass, paper, plastic, metal, and general waste. The system uses transfer learning based on the MobileNetV2 pretrained model, a lightweight and efficient deep-learning architecture developed by Google for image classification tasks. Transfer learning allows the model to leverage features already learned from the ImageNet dataset, improving accuracy while reducing training time and computational cost. The system is intended to support sustainable waste management practices by automating the sorting process and reducing human error.

A. Dataset Details

Although the original TrashNet dataset contains six waste categories, this study uses only five of them—glass, paper, plastic, metal, and general waste—to align with the scope of the proposed recycling bin system.

All images are resized to 224×224 pixels and normalized to ensure consistent input to the MobileNetV2 model. The images are organized into class-based subfolders to facilitate efficient data loading. To increase the variability of the training data and enhance the model's robustness in real-world scenarios, several

data augmentation techniques are applied. These transformations help the model generalize better by simulating different environmental conditions and variations in the appearance of waste items, ensuring the system can reliably contribute to sustainable recycling practices.

TABLE I: Data preprocessing and augmentation techniques.

Augmentation	Purpose / Effect
Rotation	Makes the model handle rotated waste items.
Width & Height Shift	Helps the model recognize objects slightly off-center.
Zoom In/Out	Simulates objects at different distances from the camera.
Horizontal Flip	Ensures the model can recognize mirrored objects.
Normalization	Scales pixel values to 0–1 for MobileNetV2 input.



(a) Paper (b) Metal (c) Glass

(d) Plastic (e) Paper (f) Plastic

(g) Metal (h) Glass (i) Waste

Fig. 4: Sample images from the dataset.

B. Model Design

MobileNetV2 is used as the base model to extract features from images. Its convolutional base is frozen to preserve the knowledge it has already learned from the large ImageNet dataset. On top of MobileNetV2, a custom classification head is added, consisting of:

- Global Average Pooling layer to summarize extracted features.
- Fully connected dense layers for classification.
- Dropout layers to reduce overfitting.
- Softmax layer to predict probabilities for each of the five waste categories.

This architecture allows the system to quickly learn to identify waste types without requiring a large dataset, while maintaining accuracy and supporting sustainable waste management.

Before training, all images are resized to 224×224 pixels and normalized to match MobileNetV2 input requirements. This preprocessing standardizes lighting and scale without removing important visual details, allowing the model to extract stable features. The final implementation also supports batch inference, enabling entire folders of images to be processed at once rather than a single image per prediction. This improves evaluation speed and aligns with real smart-bin usage where multiple items may be classified continuously.

C. Training Process

The model is trained using an 80%–20% train-test split. The training configuration includes:

- **Loss function:** Categorical Cross-Entropy
- **Epochs:** 12a (sufficient due to transfer learning)
- **Regularization:** Dropout layers and early stopping to prevent overfitting

During training, accuracy and loss metrics are monitored to ensure the model successfully captures essential features from the dataset. This setup allows the system to perform an accurate, real-time classification, contributing to efficient and sustainable recycling practices.

Results and Discussion

The model was initially trained from scratch using a standard CNN on the original six-class TrashNet dataset; however, it exhibited poor generalization, achieving only 57% validation accuracy, which is below the approximately 63% accuracy reported for the original TrashNet implementation. The approach was substituted with transfer learning using MobileNetV2 due to the limited dataset size and insufficient feature learning. This architecture, pretrained on ImageNet, offers robust low-level and mid-level feature extraction like edges, textures, and forms, minimizing the necessity to learn representations from scratch. Only the upper layers were fine-tuned for waste classification, resulting in a notable enhancement in performance and efficiency. The number of categories was decreased from six to five, enhancing class balance and model stability.

To increase real-world performance, the dataset was enhanced by integrating more images captured with a phone and collected from online sources. After retraining with the revised dataset, the validation accuracy remained within the range of 75–78%; however, the model displayed considerably better performance on unseen, real-world inputs, showing increased generalization beyond the original dataset.

TABLE II: Performance Summary of the Final MobileNetV2 Model

Metric	**Result**
Training accuracy	95%
Validation accuracy	75–78%
Epoch stopped	∼13/17 (Early stopping)
Inference speed	∼138 images/s

The final MobileNetV2 model achieved a top-1 accuracy of 78.24% and an inference speed of nearly 138 images per second, confirming its suitability for real-time smart-bin implementation. The model shows the best performance on paper (89%) and plastic (90%), although glass (71%), general waste (73%), and particularly metal (59%) present greater challenges due to increased visual ambiguity and overlap with other categories.

Variations in performance are mostly influenced by visual complexity and environmental conditions. Paper and plastic exhibit consistent texture and form, leading to improved precision in feature extraction. However, metal and glass are influenced by reflections, transparency, and similarities to surrounding items, making them more challenging to differentiate in real-world scenarios. General trash poses challenges due to its diverse and unpredictable composition, without a consistent visual pattern.

After including real-world photos from phone captured images and web sources, the model showed a significant improvement in practical performance, despite the validation accuracy remaining within the 75–78% range. This indicates improved generalization and less misclassification in actual everyday usage as instead of only in supervised testing.

Overall, the system shows strong capability for implementation across popular waste categories, while further data expansion, especially concerning metal and general waste, would likely increase consistency and adaptability in different settings.

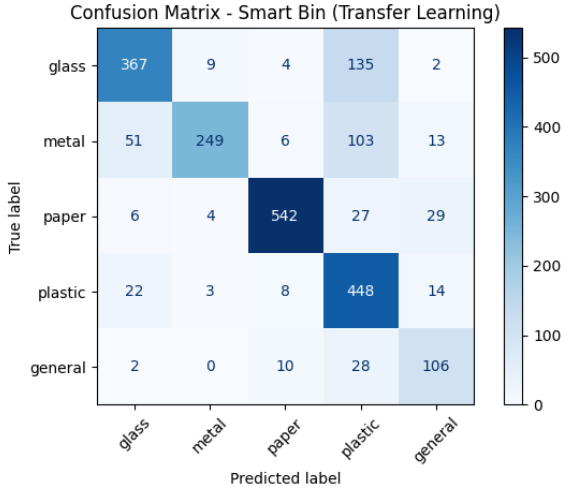


Fig. 5: Waste Classification Confusion Matrix

The model shows outstanding performance on paper and plastic, with fewer mistakes compared to other categories. The majority of misclassification occurs between metal–glass and general–plastic due to visual resemblance and uneven real-world appearances.

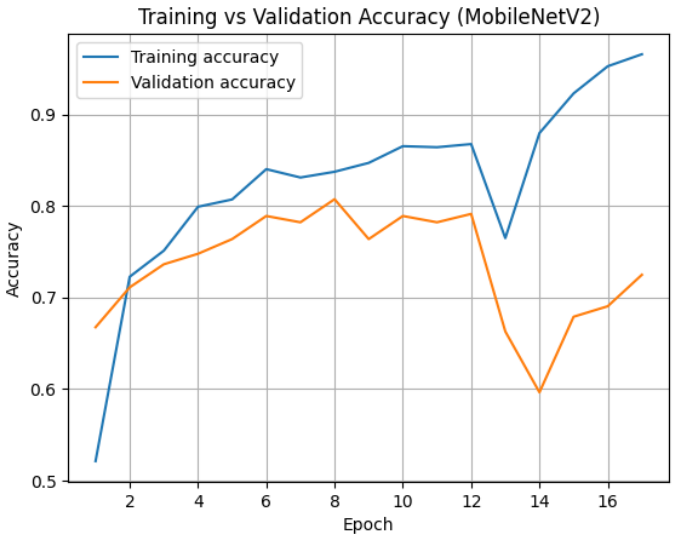


Fig. 6: Training vs Validation Accuracy (MobileNetV2)

Training accuracy increased steadily and reached above 95%, while validation accuracy peaked around 81% before dropping slightly, indicating mild overfitting in later epochs.

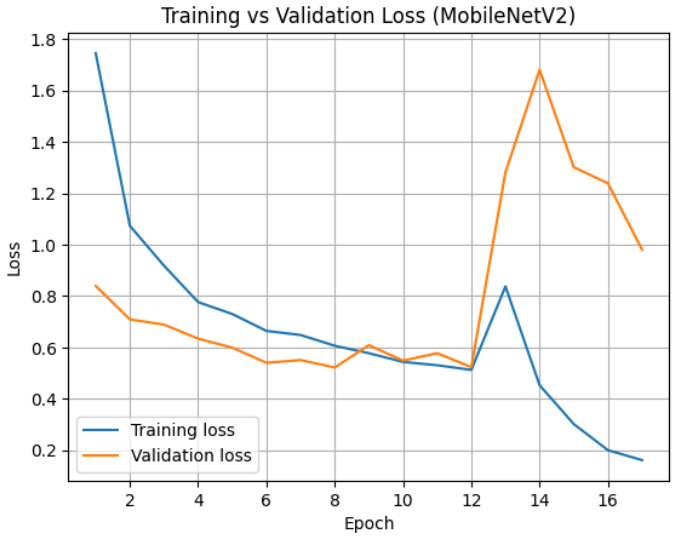


Fig. 7: Training vs Validation Loss (MobileNetV2)

Training loss consistently decreased, but validation loss rose after epoch 12, showing the model reached its optimal point earlier and began overfitting afterward.

D. Comparison With Existing Models

A baseline CNN trained on the original TrashNet dataset achieved only 57% validation accuracy, which aligns with reported TrashNet results of approximately 60–65%. Our earlier CNN implementation showed similar behavior with around 58% accuracy and weak generalization. In contrast, the proposed MobileNetV2 transfer-learning model reached

TABLE III: Performance comparison with baseline and previous models

Model	Classes	Training style	Accuracy (val / test)
Original TrashNet baseline CNN	6	Simple CNN from scratch	∼63–65% (reported)
Our first CNN model	6	Simple CNN from scratch	∼58% (validation)
Proposed MobileNetV2 model	5 + real-world images	Transfer learning + fine-tuning	78.24% (evaluated on 2,188 images)

78.24% accuracy on 2,188 test images, demonstrating a significant improvement due to pretrained feature extraction and the inclusion of real-world data.

To improve class balance and reduce label confusion, the "cardboard" category from TrashNet was eliminated, therefore decreasing the dataset from six to five classes (glass, metal, paper, plastic, general garbage). This modification enhanced the model's ability to distinguish visual limits, leading to more consistent divergence. The enhancement is mostly credited to pretrained feature extraction, less class ambiguity, and the inclusion of actual data.

These results indicate that the final model is suitable for actual smart-bin implementation rather than only controlled experimentation. The enhanced MobileNetV2 model demonstrates reliable classification performance across different lighting conditions, angles, and backgrounds, overcoming the shortcomings identified in previous versions.

A user-friendly interface was created to validate predictions on real-world photos. The interface enables users to upload one or several images and presents the projected class along with the confidence score in real time. Although it does not influence model performance, it illustrates usability for non-technical users and showcases the system's functionality in actual smart-bin applications.

Conclusion

In this research, an AI-powered recycling bin sorting system was developed to automatically classify five common waste categories: plastic, paper, glass, metal and general waste. The system was designed to address the limitations of manual sorting, such as slow processing, human error, and inconsistent recycling quality. Using transfer learning with the MobileNetV2 model, the system was able to learn from both the TrashNet dataset and additional real-world images, achieving much higher accuracy and better generalization than traditional CNN models.

The results show that the proposed model is fast, efficient, and reliable enough for real-time use. It reached a validation accuracy of about 78%, with strong performance in paper and plastic, and acceptable performance in categories that are more visually challenging such as metal and glass. The model also demonstrated robustness when tested on new images taken under different lighting conditions, angles, and levels of noise, proving its potential for real-world implementation.

Compared to earlier approaches—including CNNs baselines and existing smart-bin prototypes—the MobileNetV2 transfer-learning model offers better accuracy, faster inference speed, and improved adaptability. The inclusion of additional phone-captured images further strengthened its real-world performance, showing that even modest dataset improvements can significantly enhance generalization.

Overall, the system presents a practical and scalable solution for smart waste management. While future improvements could focus on expanding the dataset—especially for metal and general waste—the current model already demonstrates strong performance and real potential for deployment in public spaces, schools, malls, and workplaces. By reducing sorting errors and improving recycling efficiency, this AI-powered system contributes to a cleaner, more sustainable environment.

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